Matching Realizations of Edge Quotients for Co-bipartite Graphs

Research note

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Abstract

For a finite simple co-bipartite graph G on n vertices, write $P_G(x) = (-1)^n \chi_G(-x)$, where χ_G is its chromatic polynomial. When $n \geq 3$, the Bell transform of $P_G(x)/[x(x+1)]$ is exactly a reversed matching enumerator of $K_{n-2,n-1}$ with an embedded copy of the complement edges deleted. The Heilmann–Lieb theorem therefore gives only real nonpositive zeros for this transformed quotient. Bell product preservation then implies that attaching G along an edge to any ordinary τ-real graph preserves ordinary τ-realness. A bounded extension holds for clique sizes at most the larger side of a chosen bipartition of the complement. This note proves these statements, records exact finite checks, and does not assert historical novelty or unrestricted edge closure.

1 Notation and conventions

All graphs are finite and simple. A graph is **co-bipartite** if its vertex set is the union of two cliques, equivalently if its complement is bipartite. For a graph G of order n, put

$$P_G(x) = (-1)^n \chi_G(-x), \qquad R_j(x) = \prod_{i=0}^{j-1} (x+i), \qquad R_0(x) = 1.$$

The falling factorial is $(x)_j = x(x-1)\cdots(x-j+1)$, with $(x)_0 = 1$.

The **Bell transform** is the linear map $\mathcal{B} : \mathbb{R}[x] \to \mathbb{R}[z]$ determined by

$$\mathcal{B}(x^d) = T_d(z), \qquad T_d(z) = \sum_{q=0}^{d} \left\{ {d \atop q} \right\} z^q, \qquad T_0(z) = 1,$$

where the braces denote Stirling numbers of the second kind. Thus the T_d are the Touchard polynomials and

$$\sum_{d \geq 0} T_d(z) \frac{u^d}{d!} = \exp(z(e^u - 1)).$$

Throughout this note, **ordinary τ-realness** means that $\Theta_G(z) := \mathcal{B}(P_G)(z)$ has only real zeros. This is the Bell normalization; there is no factorial rescaling of its monomial coefficients. In terms of the signed convention of Brenti–Royle–Wagner [1], the exact relation is

$$\Theta_G(z) = (-1)^n \tau_{\mathrm{BRW}}(G; -z).$$

Thus real-rootedness is the same property in the two conventions. The coefficients of P_G are nonnegative, by the usual alternating-sign property of chromatic coefficients. Consequently Θ_G has nonnegative coefficients, so its real zeros, if all its zeros are real, are nonpositive. Constant nonzero polynomials are regarded as real-rooted.

For a bipartite graph J with specified part sizes $a \leq b$, let $m_\ell(J)$ be its number of matchings of size ℓ, including $m_0(J) = 1$. Define the reversed matching enumerator by

$$M_J(z) = \sum_{\ell=0}^{a} m_\ell(J) z^{a-\ell}.$$

Its exponent normalization uses the size a of the specified first part, not the matching number. Isolated vertices remain part of the graph.

2 Exact edge quotient realization

Theorem 1. Let G be a co-bipartite graph on $n \geq 3$ vertices, and write $H = \overline{G}$. Choose a bipartition $A \sqcup B$ of H with $1 \leq |A| \leq |B|$. If H is edgeless, choose $|A| = 1$ and $|B| = n - 1$. Embed H in $K_{n-2,n-1}$ with A in the first part and B in the second, adding unused vertices to fill the parts, and set

$$J = K_{n-2,n-1} \setminus E(H).$$

Here only the embedded edges of H are deleted. Then

$$\boxed{\mathcal{B}\left(\frac{P_G}{R_2}\right)(z) = M_J(z).}$$

In particular, this transformed quotient has only real nonpositive zeros. For the remaining edge-containing case $n = 2$, necessarily $G = K_2$, and the quotient is the constant 1.

The embedding is possible because

$$|A| \leq \lfloor n/2 \rfloor \leq n - 2, \qquad |B| \leq n - 1.$$

When H has edges, both parts are nonempty; isolated complement vertices may be assigned subject to the stated part-size conditions. The assertion does not require G to be connected or to have no isolated vertices.

2.1 Chromatic expansion by complement matchings

Every independent set in G has at most two vertices. A partition of $V(G)$ into independent blocks is therefore uniquely specified by a matching of H: its edges give the two-element blocks, and its unmatched vertices give singleton blocks. A matching of size k gives $n - k$ blocks. Assigning distinct colors to these blocks yields

$$\chi_G(x) = \sum_{k \geq 0} m_k(H)(x)_{n-k}, \qquad P_G(x) = \sum_{k \geq 0} (-1)^k m_k(H) R_{n-k}(x).$$

The sign follows from $(-x)_{n-k} = (-1)^{n-k} R_{n-k}(x)$. Since $k \leq \lfloor n/2 \rfloor \leq n - 2$, every summand in the last expression is divisible by R_2.

2.2 A Bell transform identity

For integers $c \geq 1$ and $r \geq 0$, with the empty product interpreted as 1, one has

$$\mathcal{B}\left(\prod_{i=0}^{r-1}(x + c + i)\right)(z) = \sum_{\ell=0}^{r} \binom{r}{\ell} \binom{r + c - 1}{\ell} \ell!\, z^{r-\ell}.$$

To prove this identity, take the exponential generating function in r. All operations below are formal power-series operations, so no analytic convergence assumption is needed. The rising-factorial generating function and the Touchard identity give

$$\begin{aligned} \sum_{r \geq 0} \mathcal{B}\left(\prod_{i=0}^{r-1}(x + c + i)\right) \frac{u^r}{r!} &= \mathcal{B}\big((1 - u)^{-x-c}\big) \\ &= (1 - u)^{-c} \exp\Big(\frac{zu}{1 - u}\Big). \end{aligned}$$

The coefficient of $z^d u^r$ on the right is $\binom{c+r-1}{r-d}/d!$ for $0 \le d \le r$. Multiplying by $r!$ and setting $\ell = r-d$ proves the formula.

The complete bipartite graph has matching counts

$$m_\ell(K_{a,b}) = \binom{a}{\ell}\binom{b}{\ell}\ell!.$$

Hence, with $a = n-2$, $b = n-1$, $c = 2$, and $r = a-k$, the formula implies

$$\mathcal{B}\left(\frac{P_G}{R_2}\right) = \sum_{k\ge 0}(-1)^k m_k(H) M_{K_{a-k,b-k}}.$$

Each enumerator on the right is normalized to first-part size $a-k$.

2.3 Forbidden edge inclusion and exclusion

Count the ℓ-edge matchings of $K_{a,b}$ that avoid the embedded edges of H. In inclusion–exclusion, a specified set of forbidden edges can occur in such a matching only if it is itself a matching. A forbidden matching of size k leaves a complete bipartite graph $K_{a-k,b-k}$ on the unused vertices. Therefore

$$m_\ell(J) = \sum_{k=0}^{\ell}(-1)^k m_k(H) m_{\ell-k}(K_{a-k,b-k}).$$

Multiplying by $z^{a-\ell}$ and summing over ℓ gives

$$M_J(z) = \sum_{k\ge 0}(-1)^k m_k(H) M_{K_{a-k,b-k}}(z).$$

Indeed, writing $q = \ell - k$ changes the exponent to $a-k-q$, exactly the exponent in the smaller-board enumerator. There is no additional power of z. Comparison with the preceding Bell transform proves the claimed polynomial identity.

3 Real zeros and universal edge attachment

The usual matching polynomial of a graph J of order N is

$$\mu_J(y) = \sum_{\ell\ge 0}(-1)^\ell m_\ell(J) y^{N-2\ell}.$$

The Heilmann–Lieb theorem states that this polynomial has only real zeros [2]. In the edge construction, $N = 2a+1$, and direct substitution gives

$$\mu_J(y) = (-1)^a y M_J(-y^2).$$

If $z_0 \neq 0$ is a zero of M_J, choose a complex number y_0 with $y_0^2 = -z_0$. Then $y_0 \neq 0$ and $\mu_J(y_0) = 0$, so y_0 is real and $z_0 = -y_0^2 < 0$. Zero roots of M_J are allowed. This finishes the root assertion of Theorem 1. For $G = K_2$, one has $P_G = R_2$ directly.

For polynomials f and g, define

$$(f \star g)(z) = \sum_{k\ge 0}\frac{z^k}{k!} f^{(k)}(z) g^{(k)}(z),$$

a finite sum. The Bell product identity is

$$\mathcal{B}(pq) = \mathcal{B}(p) \star \mathcal{B}(q).$$

It follows, for example, by applying the Touchard generating function to $p = e^{ux}$ and $q = e^{vx}$ formally and comparing coefficients of u and v. We use the accompanying preservation theorem: if f and g have nonnegative coefficients and only real nonpositive zeros, then so does $f \star g$. This is the Bell-normalized form of Brenti–Royle–Wagner Proposition 5.4 [1].

Corollary 2. Let G be an edge-containing co-bipartite graph, and let F be any ordinary τ-real graph. Suppose their intersection consists of exactly the two vertices and the edge of a common K_2. Form their union with no other vertex identifications and no additional edges between their remaining vertices. Then $G \cup F$ is ordinary τ-real.

Proof. The chromatic clique-sum identity [1] gives

$$\chi_{G\cup F}(x) = \frac{\chi_G(x)\chi_F(x)}{(x)_2}, \qquad P_{G\cup F}(x) = \frac{P_G(x)P_F(x)}{R_2(x)}.$$

Consequently

$$\Theta_{G\cup F} = \mathcal{B}(P_G/R_2) \star \Theta_F.$$

Both factors have nonnegative coefficients and only real nonpositive zeros, by Theorem 1 and the definition of ordinary τ-realness. Product preservation proves the claim. Taking $F = K_2$, for which $\Theta_F(z) = z^2 + 2z$, also shows that G itself is ordinary τ-real. □

4 A bounded extension to larger cliques

Theorem 3. Retain $n \geq 3$ and a chosen complement bipartition $A \sqcup B$ as in Theorem 1, and write $s = |A| \leq t = |B|$. For every integer j with $1 \leq j \leq t$, set

$$a = n - j, \qquad b = n - 1, \qquad J_j = K_{a,b} \setminus E(H),$$

embedding A in the first part and B in the second. Then

$$\mathcal{B}\left(\frac{P_G}{R_j}\right)(z) = M_{J_j}(z), \qquad \mu_{J_j}(y) = (-1)^{n-j} y^{j-1} M_{J_j}(-y^2).$$

In particular, the transformed quotient has only real nonpositive zeros, and attachment of G along a j-clique to any ordinary τ-real graph preserves ordinary τ-realness, with the same restrictions on the intersection and union as in Corollary 2.

Proof. The bound $j \leq t$ implies $s \leq n - j = a$, while $t \leq n - 1 = b$. Thus the required embedding exists. Every complement matching has size $k \leq s \leq a$, so $n - k \geq j$ and every term R_{n-k} in the chromatic expansion is divisible by R_j. Apply the Bell identity of Section 2.2 with $c = j$ and $r = n - j - k = a - k$; then $r + c - 1 = b - k$. The same inclusion–exclusion proof yields M_{J_j}. Since J_j has $a + b = 2a + j - 1$ vertices, its matching polynomial has the displayed sign and factor y^{j-1}. The Heilmann–Lieb argument again gives the zeros. Finally, for a common j-clique the chromatic identity gives $P_{G\cup F} = P_G P_F / R_j$, and Bell product preservation applies. □

The bound refers to the larger side of the **chosen complement bipartition**, not to an arbitrary clique size of G. A clique containing vertices from both sides may have more than t vertices. No attachment claim for those larger intersections follows from this embedding proof. The $n = 2$ edge boundary is handled separately in Theorem 1, not by this bounded extension.

5 An infinite family without join or clique separation

A **nontrivial join decomposition** is a partition into two nonempty vertex sets with every possible edge between them. A **clique separator** is a clique whose deletion leaves a disconnected graph.

Corollary 4. For every $m \geq 4$, the graph $G = \overline{C_{2m}}$ satisfies Theorem 1 and has neither a nontrivial join decomposition nor a clique separator.

Proof. The even cycle C_{2m} is bipartite, so G is co-bipartite. Its complement is connected, excluding a nontrivial join decomposition. Suppose that a clique K of G disconnects G when deleted. The set K is independent in C_{2m}, so $|K| \leq m$ and at least $m \geq 4$ vertices remain. Distinct components of $G - K$ have all possible cross edges in C_{2m}. Three or more components would therefore give a triangle in the cycle, which is impossible. If there are exactly two components with vertex sets X, Y, the cycle's maximum degree 2 forces $|X|, |Y| \leq 2$. At least four vertices remain, so both sets have size 2, and their cross edges give a C_4 subgraph of C_{2m}. A cycle of length at least 8 contains no such subgraph. This contradiction proves the claim. □

This corollary excludes these two decomposition mechanisms for the stated family; it does not assert exclusion from every earlier closure family.

6 Exact verification

The accompanying verifier performs exact integer polynomial and matching computations, together with rational real-root isolation. From the directory containing the supplied verification files, run

```
python verify.py
```

The certificate records **165 exact cases**: all 147 edge-containing co-bipartite graphs in the graph atlas through seven vertices, and 18 deterministic examples on 8–12 vertices obtained as complements of paths, even cycles, double stars, and complete bipartite graphs. For $n \geq 3$, three constructions agree: stable-partition chromatic computation followed by the Bell transform, direct matching dynamic programming on the allowed board, and forbidden-board matching inclusion–exclusion. The $n = 2$ case is a separate direct check of $G = K_2$ and quotient 1.

Across the $n \geq 3$ records, a further **663 bounded-j quotient identities and exact root checks** pass for the recorded complement bipartitions. These computations support the identities and help detect normalization or implementation errors; they are not premises of the infinite-family proofs.

7 Scope and provenance

The graph-specific mechanism has two parts: complement matchings encode all independent-set partitions, and the bipartite complement embeds in a rectangular matching board. Independent sets of size at least three destroy the matching encoding. A triangle-free but nonbipartite complement still gives the encoding, but not the required board embedding. The argument therefore does not establish unrestricted ordinary edge closure.

Brenti–Royle–Wagner Proposition 5.12 concerns intersections of at most one vertex; Proposition 5.13 gives a different chromatic-root sufficient criterion for clique attachment [1]. This note makes no claim that its examples fail that criterion, that this class strictly extends every earlier closure class, or that the matching realization is absent from the literature. Historical novelty remains unknown, and overlap with Mu–Shen–Xu remains unassessed.

This research note is dated 2026-11-02, and its proof was developed with AI assistance. A separate, independent AI review checked the matching correspondence, signs, Bell identity, inclusion–exclusion

exponents, embedding bounds, matching-polynomial conversion, and bounded clique extension, without finding a mathematical correction to the main identity. That review is not human peer review or proof-assistant verification.

References

[1] Francesco Brenti, Gordon F. Royle, and David G. Wagner. *Location of Zeros of Chromatic and Related Polynomials of Graphs.* Canadian Journal of Mathematics **46**(1), 55–80 (1994). doi:10.4153/CJM-1994-002-3.

[2] Ole J. Heilmann and Elliott H. Lieb. *Theory of monomer-dimer systems.* Communications in Mathematical Physics **25**, 190–232 (1972). doi:10.1007/BF01877590.